

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2.	(a)			× (1) for each of the following: Box 2 All em waves travel at the same speed in a vacuum. Box 3 Gamma rays can be emitted from radioactive materials. Box 4 The wavelength of em waves decrease from radio waves to gamma rays. Any extra ticks deduct one mark for each extra tick.	3			3		
	(b)	(i)	I	10 [m]	1			1		
			II	2 [waves]		1		1		
			III	3 [m]		1		1		
		(ii)	I	7.5 Hz (1) second (1)	2			2		
			II	Substitution: 3 (ecf) × 7.5 (1) = 22.5 [m/s] (1)	1	1		2	2	
	(c)			Award 1 mark for more waves or smaller wavelength Award 2 marks for twice the number of waves (4 waves) or half the wavelength (1.5 m) [so student correct] Accept for 1 mark the distance between the waves is <u>halved</u>			2	2	1	
				Question 2 total	7	3	2	12	3	0